FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

FLAB3

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Name of Student	Name of Professor
Oct 29, 2026	Oct 29, 2026
Data Performed	Date Submitted

Direction: COR Program, Input, Output. Implement MACRO and PROCEDURE. Compute for the Tuition Fee.

Explanation of the Program:

The program uses macros to simplify repeated input and output operations through READ, PRINT, PRINTBUF, and EXIT. The READ macro displays each prompt and stores the user's input in its own buffer. After collecting the information, the PRINT and PRINTBUF macros display the enrollment form and course schedule in a formatted table. The ascii to integer procedure converts the entered number of units from ASCII characters into an integer. The compute tuition procedure multiplies the units by 1,000, while integer_to_ascii converts the computed tuition back into printable characters. Finally, the macros display the tuition rate and the formatted Total Tuition Fee as part of the desired output.

Output Reference (COR):

Term - School Year : 1-20262027 Submit

Enrollment Status : ENROLLED
Student # : 202511495
Name : BERGONIA, JOHAN JB AWINGAN
Phone : +630961448784 Zip Code : 1850
Classification :

College : COMPUTER STUDIES Program : BSCSSE
Year Level : 2 1st term, SY 26-27
Address : 16 IDAHO STREET, PHASE 2 GREENLAND NEWTOWN, BANABA

Courses	Title	Section	Units	Days	Time	Room
CCS0021	INFORMATION MANAGEMENT (LEC)	TN24	2	T	16:00:00-18:40:00	ONLINE
CCS0021L	INFORMATION MANAGEMENT (LAB)	TN24	1	F	16:00:00-18:50:00	ONLINE
CS0001	DISCRETE STRUCTURES 1	TN24	3	S / W	15:00:00-16:50:00 / 15:00:00-16:50:00	F710 / F710
CS0003	COMPUTER SYSTEMS & ARCHITECTURE	TN24	2	F	10:00:00-12:40:00	ONLINE
CS0003L	COMPUTER SYSTEMS & ARCHITECTURE (LAB)	TN24	1	TH	10:00:00-12:50:00	E610
CS0070	OBJECT ORIENTED PROGRAMMING	TN24	2	F	07:00:00-09:40:00	ONLINE
CS0070L	OBJECT ORIENTED PROGRAMMING-LAB	TN24	1	TH	07:00:00-09:50:00	E609
GED0021	SPECIALIZED ENGLISH PROGRAM 2	TN24	3	S / W	11:00:00-12:50:00 / 11:00:00-12:50:00	F807 / F807
GED0075	LINEAR ALGEBRA	TN24	3	F / T	13:00:00-14:50:00 / 13:00:00-14:50:00	ONLINE / ONLINE
GED0081	COLLEGE PHYSICS 1 LECTURE	TN24	2	T	07:00:00-09:40:00	ONLINE
GED0081L	COLLEGE PHYSICS 1 LABORATORY	TN24	1	S	07:00:00-09:50:00	F1003
TOTAL UNITS 21						

Program Compilation:

```
Problems  Output  Debug Console  Terminal  Ports

@johanjbbergonia-source ➜ /workspaces/2Y1T-CS0003L-FLAB3 (main) $ nasm -f elf32 FLAB3.asm -o FLAB3.o
@johanjbbergonia-source ➜ /workspaces/2Y1T-CS0003L-FLAB3 (main) $ ld -m elf_i386 FLAB3.o -o FLAB3
@johanjbbergonia-source ➜ /workspaces/2Y1T-CS0003L-FLAB3 (main) $ ./FLAB3
```

Output Screenshots:

```
Enrollment Status: ENROLLED
Student #: 202511495
College: COMPUTER STUDIES
Program: BSCSSE
Name: BERGONIA, JOHAN JB AWINGAN
Year Level: 2    1st term, SY 26-27
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Address: 16 IDAHO STREET, PHASE 2 GREENLAND NEWTOWN, BANABA

============================================================
                    ENROLLMENT FORM
============================================================
Enrollment Status: ENROLLED
Student #: 202511495
Name: BERGONIA, JOHAN JB AWINGAN
College: COMPUTER STUDIES
Program: BSCSSE
Year Level: 2    1st term, SY 26-27
Phone: +630961448784
Zip Code: 1850
Address: 16 IDAHO STREET, PHASE 2 GREENLAND NEWTOWN, BANABA

+---------+-----------------------------------------+--------+------+------+-----------------------------------------+----------------+
| COURSE  | TITLE                                   | SECTION| UNITS| DAYS | TIME                                    | ROOM           |
+---------+-----------------------------------------+--------+------+------+-----------------------------------------+----------------+
| CCS0021 | INFORMATION MANAGEMENT (LEC)            | TN24   | 2    | T    | 16:00:00-18:40:00                       | ONLINE         |
| CCS0021L| INFORMATION MANAGEMENT (LAB)            | TN24   | 1    | F    | 16:00:00-18:50:00                       | ONLINE         |
| CS0001  | DISCRETE STRUCTURES 1                   | TN24   | 3    | S / W| 15:00:00-16:50:00 / 15:00:00-18:50:00   | F710 / F710    |
| CS0003  | COMPUTER SYSTEMS & ARCHITECTURE         | TN24   | 2    | F    | 10:00:00-12:40:00                       | ONLINE         |
| CS0003L | COMPUTER SYSTEMS & ARCHITECTURE (LAB)   | TN24   | 1    | TH   | 10:00:00-12:50:00                       | E610           |
| CS0070  | OBJECT ORIENTED PROGRAMMING             | TN24   | 2    | F    | 07:00:00-09:40:00                       | ONLINE         |
| CS0070L | OBJECT ORIENTED PROGRAMMING-LAB         | TN24   | 1    | TH   | 07:00:00-09:50:00                       | E609           |
| GED0021 | SPECIALIZED ENGLISH PROGRAM 2           | TN24   | 3    | S / W| 11:00:00-12:50:00 / 11:00:00-12:50:00   | F607 / F607    |
| GED0075 | LINEAR ALGEBRA                          | TN24   | 3    | F / T| 13:00:00-14:50:00 / 13:00:00-14:50:00   | ONLINE / ONLINE|
| GED0081 | COLLEGE PHYSICS 1 LECTURE               | TN24   | 2    | T    | 07:00:00-09:40:00                       | ONLINE         |
| GED0081L| COLLEGE PHYSICS 1 LABORATORY            | TN24   | 1    | S    | 07:00:00-09:50:00                       | F1003          |
+---------+-----------------------------------------+--------+------+------+-----------------------------------------+----------------+
| TOTAL UNITS: 21                                                                                                    |                |
+---------+-----------------------------------------+--------+------+------+-----------------------------------------+----------------+

Enter Total No. of Units: 21

Tuition Fee Price: Php 1000 per unit
Total Tuition Fee: Php 21000
@johanjbbergonia-source ➜ /workspaces/2Y1T-CS0003L-FLAB3 (main) $
```

Source Code:

```
%macro PRINT 1
    mov eax, 4
    mov ebx, 1
    mov ecx, %1
    mov edx, %1_end - %1
    int 80h
%endmacro

%macro WRITE 2
    mov eax, 4
    mov ebx, 1
    mov ecx, %1
    mov edx, %2
    int 80h
%endmacro

%macro PRINTBUF 1
```

```
    mov ecx, %1
    xor edx, edx
%%find_end:
    cmp byte [ecx + edx], 0
    je %%write_buffer
    inc edx
    jmp %%find_end
%%write_buffer:
    mov eax, 4
    mov ebx, 1
    int 80h
%endmacro

%macro READ 3
    PRINT %1
    mov eax, 3
    mov ebx, 0
    mov ecx, %2
    mov edx, %3
    int 80h
    cmp eax, 0
    jle %%read_done
    mov edx, eax
    dec edx
    cmp byte [%2 + edx], 10
    je %%remove_newline
    cmp byte [%2 + edx], 13
    jne %%terminate
%%remove_newline:
    mov byte [%2 + edx], 0
    jmp %%read_done
%%terminate:
    mov byte [%2 + eax], 0
%%read_done:
%endmacro

%macro EXIT 0
    mov eax, 1
    xor ebx, ebx
    int 80h
%endmacro

section .data
```

```
prompt_status db "Enrollment Status: ", 0
prompt_status_end:
prompt_student db "Student #: ", 0
prompt_student_end:
prompt_college db "College: ", 0
prompt_college_end:
prompt_program db "Program: ", 0
prompt_program_end:
prompt_name db "Name: ", 0
prompt_name_end:
prompt_year db "Year Level: ", 0
prompt_year_end:
prompt_phone db "Phone: ", 0
prompt_phone_end:
prompt_zip db "Zip Code: ", 0
prompt_zip_end:
prompt_address db "Address: ", 0
prompt_address_end:
prompt_units db 10, "Enter Total No. of Units: ", 0
prompt_units_end:

title db 10,
"=============================================================",
10
    db "                ENROLLMENT FORM", 10
    db
"=============================================================",
10, 0
title_end:

label_status db "Enrollment Status: ", 0
label_status_end:
label_student db "Student #: ", 0
label_student_end:
label_college db "College: ", 0
label_college_end:
label_program db "Program: ", 0
label_program_end:
label_name db "Name: ", 0
label_name_end:
label_year db "Year Level: ", 0
label_year_end:
```

```
label_phone db "Phone: ", 0
label_phone_end:
label_zip db "Zip Code: ", 0
label_zip_end:
label_address db "Address: ", 0
label_address_end:

course_table db 10, "+---------+-------------------------------------------+--------+------+-------+-----------------------------------+---------------+", 10
            db "| COURSE  | TITLE                                     | SECTION| UNITS| DAYS | TIME                              | ROOM          |", 10
            db "+---------+-------------------------------------------+--------+------+-------+-----------------------------------+---------------+", 10
            db "| CCS0021 | INFORMATION MANAGEMENT (LEC)              | TN24   | 2    | T    | 16:00:00-18:40:00                 | ONLINE        |", 10
            db "| CCS0021L| INFORMATION MANAGEMENT (LAB)              | TN24   | 1    | F    | 16:00:00-18:50:00                 | ONLINE        |", 10
            db "| CS0001  | DISCRETE STRUCTURES 1                     | TN24   | 3    | S / W| 15:00:00-16:50:00 / 15:00:00-18:50:00    | F710 / F710   |", 10
            db "| CS0003  | COMPUTER SYSTEMS & ARCHITECTURE           | TN24   | 2    | F    | 10:00:00-12:40:00                 | ONLINE        |", 10
            db "| CS0003L | COMPUTER SYSTEMS & ARCHITECTURE (LAB)     | TN24   | 1    | TH   | 10:00:00-12:50:00                 | E610          |", 10
            db "| CS0070  | OBJECT ORIENTED PROGRAMMING               | TN24   | 2    | F    | 07:00:00-09:40:00                 | ONLINE        |", 10
            db "| CS0070L | OBJECT ORIENTED PROGRAMMING-LAB           | TN24   | 1    | TH   | 07:00:00-09:50:00                 | E609          |", 10
            db "| GED0021 | SPECIALIZED ENGLISH PROGRAM 2             | TN24   | 3    | S / W| 11:00:00-12:50:00 / 11:00:00-12:50:00    | F607 / F607   |", 10
            db "| GED0075 | LINEAR ALGEBRA                            | TN24   | 3    | F / T| 13:00:00-14:50:00 / 13:00:00-14:50:00    | ONLINE / ONLINE|", 10
            db "| GED0081 | COLLEGE PHYSICS 1 LECTURE                 | TN24   | 2    | T    | 07:00:00-09:40:00                 | ONLINE        |", 10
            db "| GED0081L| COLLEGE PHYSICS 1 LABORATORY              | TN24   | 1    | S    | 07:00:00-09:50:00                 | F1003         |", 10
            db "+---------+-------------------------------------------+--------+------+-------+-----------------------------------+---------------+", 10
            db "| TOTAL UNITS: 21
|             |", 10
            db "+---------+-------------------------------------------+--------+------+-------+-----------------------------------+---------------+", 10, 0
course_table_end:
```

```
tuition_rate db 10, "Tuition Fee Price: Php 1000 per unit", 10, 0
tuition_rate_end:
tuition_total db "Total Tuition Fee: Php ", 0
tuition_total_end:
newline db 10, 0
newline_end:

section .bss

status resb 64
student resb 64
college resb 64
program resb 64
student_name resb 128
year_level resb 32
phone resb 64
zip_code resb 32
address resb 256
units_text resb 32
tuition_text resb 32

section .text
global _start

_start:
    READ prompt_status, status, 63
    READ prompt_student, student, 63
    READ prompt_college, college, 63
    READ prompt_program, program, 63
    READ prompt_name, student_name, 127
    READ prompt_year, year_level, 31
    READ prompt_phone, phone, 63
    READ prompt_zip, zip_code, 31
    READ prompt_address, address, 255

    PRINT title
    PRINT label_status
    PRINTBUF status
    PRINT newline
    PRINT label_student
    PRINTBUF student
    PRINT newline
    PRINT label_name
```

```
    PRINTBUF student_name
    PRINT newline
    PRINT label_college
    PRINTBUF college
    PRINT newline
    PRINT label_program
    PRINTBUF program
    PRINT newline
    PRINT label_year
    PRINTBUF year_level
    PRINT newline
    PRINT label_phone
    PRINTBUF phone
    PRINT newline
    PRINT label_zip
    PRINTBUF zip_code
    PRINT newline
    PRINT label_address
    PRINTBUF address
    PRINT newline
    PRINT course_table

    READ prompt_units, units_text, 31
    mov esi, units_text
    call ascii_to_integer
    push eax
    call compute_tuition
    mov edi, tuition_text
    call integer_to_ascii
    pop eax

    PRINT tuition_rate
    PRINT tuition_total
    PRINTBUF tuition_text
    PRINT newline
    EXIT

ascii_to_integer:
    push ebx
    push ecx
    push edx
    xor eax, eax
    xor ecx, ecx
```

```
.convert_loop:
    mov bl, [esi + ecx]
    cmp bl, 0
    je .convert_done
    imul eax, 10
    sub bl, '0'
    movzx edx, bl
    add eax, edx
    inc ecx
    jmp .convert_loop
.convert_done:
    pop edx
    pop ecx
    pop ebx
    ret

compute_tuition:
    imul eax, 1000
    ret

integer_to_ascii:
    push ebx
    push ecx
    push edx
    cmp eax, 0
    jne .build_number
    mov byte [edi], '0'
    mov byte [edi + 1], 0
    jmp .ascii_done
.build_number:
    mov ebx, 10
    xor ecx, ecx
.divide_loop:
    xor edx, edx
    div ebx
    push edx
    inc ecx
    cmp eax, 0
    jne .divide_loop
.write_loop:
    pop eax
    add al, '0'
    mov [edi], al
```

```
    inc edi
    loop .write_loop
    mov byte [edi], 0
.ascii_done:
    pop edx
    pop ecx
    pop ebx
    ret
```